

CS 5330 HW 12

1. How do the Gatys and Johnson style transfer papers define content?

Content is what features, objects, details the image actually includes. It is described mathematically by the activations of the network

2. How do the Gatys and Johnson style transfer papers define style?

Style is about the relationship between channels. It exists at all levels of the image (fine detail to high level detail). Specifically the Gram matrix is the mathematical descriptor of style

3. How does a Generative Adversarial Network solve the problem of training a generator to make images that are 'like' a set of training images without training the network to generate those exact training images?

The purpose of the generator (synthesis network) is not to produce the exact training images, but ones that are similar. It does this by using a discriminator network which predicts whether the image is real or fake. The goal is to "fool" that network, and the synthesizer network is fine tuned to minimize "adversarial loss" through backpropagation until that is achieved

4. What is a common failure mode of GANs?

There are two networks to train, and the discriminator "pulls along" the synthesizer. It is hard to balance training the discriminator vs training the synthesizer

5. What was the innovation of the Style-GAN paper over a vanilla GAN architecture?

An alternative synthesis network that takes noise, runs it through a fully connected MLP and returns a vector w . The argument is that w is easier to decode than the original noise, and that this network effectively transforms or disentangles the latent space qualities.

6. How does the Creative-Net paper define novelty in art?

Novelty is something that is not easily classifiable as an *existing* art style.

7. A diffusion network learns to estimate what property of an input image?

It learns the “noise field” of the image, which when subtracted from the pre-network noise image, results in a (slightly) de-noised the image. This is done many many times before arriving at the final image.

8. How do you create input/output pairs for training a diffusion network?

In training a diffusion network, noise is iteratively added to an image over many steps, resulting a string of a progressively more noisy version of that image. At any point in the chain of images, we can sample an input (before noise), and the output (after noise) images, along with the noise that generated the transformation. These three things together are at the core of diffusion network training.

9. When generating a new image, what is the process a diffusion network follows?

GAN could succeed at reducing the loss but also fail at producing good images, due to things like mode collapse (only producing one high quality output)